

Approved

State of Idaho
DOPL

PA#: 1906305
Date: 11/20/2025

These plans are approved contingent on the compliance with the mark-ups and notes applied.

This approval shall not be construed to be an approval of any violation of, or variance from Idaho's adopted codes, standards, laws or rules applicable to this project.



DOPL

DIVISION OF OCCUPATIONAL & PROFESSIONAL LICENSES

VEHICLE EXHAUST GAS DETECTION SYSTEM SCHEDULE (SHOP)

SYMBOL	AREA SERVED	UNIT TYPE	GAS DETECTION RANGE			ELECTRICAL		MANUFACTURER AND MODEL	REMARKS
			CO (PPM)	NO2 (PPM)	COMBUSTIBLE (%LEL)	SENSORS	CONTROL PANEL		
			LOW / HIGH	LOW / HIGH	LOW / HIGH	VOLTS	V/Ø		
EF-1	STORAGE / REPAIR BAYS								
EF-2 & SF-1			25 - 200	1 - 5	10% - 20%	24V	120/1	MACURCO CONTROL PANEL MODEL: DVP-120 MACURCO SENSOR MODEL: CX-6	1, 2, 3

- APPROVED ALTERNATE MANUFACTURERS: ARMSTRONG AND HONEYWELL.
- PROVIDE WITH MANUFACTURER CONTROL PANEL: 3 10AMP RELAYS, 2 HORNS & 2 STROBES DRIVER, 12 ANALOG (DVP-120). PANEL INCLUDES: TIMED DAY SELECTIONS, ALARM, WARNING, AND TROUBLE INDICATIONS.
- INSTALL MANUFACTURER REQUIRED SENSORS TO COVER ENTIRE SPACE. MANUFACTURER TO SUBMIT SENSOR LAYOUT FOR FINAL APPROVAL.

FAN SCHEDULE

SYMBOL	AREA SERVED	UNIT TYPE	BLOWER				ELECTRICAL		MAXIMUM SONES	OPERATING WEIGHT (LBS)	MANUFACTURER AND MODEL	REMARKS
			CFM	ESP	MAXIMUM RPM	DRIVE	HP/W	V/Ø				
<u>EF-1</u>	STORAGE / REPAIR BAY	WALL PROP	185	0.25	1,725	DIRECT / ECM	1/8 HP	115/1	6.1	75	COOK MODEL 10XW24D17	1, 3, 5
<u>EF-2</u>	STORAGE / REPAIR BAY	TUBE AXIAL	2,800	0.25	1,725	DIRECT	1.0 HP	208/3	21	150	COOK MODEL 20D17	1, 2, 4, 6
<u>SF-1</u>	STORAGE / REPAIR BAY	TUBE AXIAL	2,800	0.25	1,725	DIRECT	1.0 HP	208/3	21	150	COOK MODEL 20D17	1, 2, 4, 7

REMARKS:

- APPROVED ALTERNATE MANUFACTURERS: ACME, GREENHECK, PENNBARRY, TWIN CITY FAN COMPANY, AND SOLER & PALAU.
- PROVIDE UNIT WITH MANUFACTURER'S FAN MOUNTED VFD (200-240/1 IN – 200-240/3 OUT), NEMA 3 PRE-WIRED DISCONNECT, INLET GUARD AND HANGING BRACKETS.
- PROVIDE WITH MANUFACTURER'S FAN MOUNTED SPEED CONTROLLER, NEMA 1 PRE-WIRED DISCONNECT, WALL COLLAR, WIREGUARD ON MOTOR SIDE, AND WEATHER HOOD.
- CONTROL FAN WITH NO₂ AND CO SENSORS. SEE VEHICLE EXHAUST GAS DETECTION SYSTEM SCHEDULE.
- FAN TO RUN CONTINUOUSLY.
- PROVIDE WITH UPBLAST DISCHARGE.
- PROVIDE WITH DOWNBLAST DISCHARGE.

DUCTLESS SPLIT HIGH WALL COOLING & HEATING UNIT SCHEDULE

SYMBOL	AREA SERVED	NOMINAL TONS	UNIT TYPE	SUPPLY FAN			COOLING REQUIRED AT 95°F OSA, 80°F EDB, 62°F EWB		HEATING REQUIRED AT 0°F OSA, 69°F EDB.	ELECTRICAL OUTDOOR UNIT			MINIMUM SEER2 / HSPF2	INDOOR/ OUTDOOR OPERATING WEIGHT (LBS)	MANUFACTURER AND MODEL	REMARKS
				CFM	HP	V/Ø	TOTAL MBH	SENSIBLE MBH	TOTAL MBH	MCA	MOCP	V/Ø				
FC-1, HP-1	OFFICE	0.5	HIGH WALL COOL/HEAT UNIT	147-383	.28	THROUGH OUTDOOR UNIT	11.3	10.8	6.40	12	15	208/1	25.7/14.8	25/65	CARRIER INDOOR UNIT MODEL 45MAHAQ06 CARRIER OUTDOOR UNIT MODEL 37MAHAQ06	1, 2, 3, 4, 5, 6
FC-2, HP-2	CONFERENCE ROOM	1.5	HIGH WALL COOL/HEAT UNIT	353-559	.037	THROUGH OUTDOOR UNIT	17.1	11.3	15.50	19	20	208/1	21.5/14.8	35/75	CARRIER INDOOR UNIT MODEL 45MAHAQ18 CARRIER OUTDOOR UNIT MODEL 37MAHAQ18	1, 2, 3, 4, 5, 6

REMARKS:

- APPROVED ALTERNATE MANUFACTURERS: LENNOX, MITSUBISHI, PANASONIC, SAMSUNG, LG, DAIKIN , OR APPROVED EQUAL BY ENGINEER.
- CONTROL UNIT WITH MANUFACTURER'S HARD-WIRED WALL MOUNTED 7 DAY PROGRAMMABLE THERMOSTAT WITH AUTO CHANGE OVER.
- PROVIDE MANUFACTURERS CRANKCASE HEATER , LOW AMBIENT CONTROLS & (TO -0°F COOLING TO -0°F HEATING) WIND BAFFLES, REFRIGERATION LINE SET SIZED BY MANUFACTURER, AND TAMPER PROOF PORT CAPS.
- PROVIDE WITH MIRO INDUSTRIES HEAVY DUTY MECHANICAL GALVANIZED SUPPORT WITH ADJUSTABLE SUPPORT LEGS. SUPPORT SHALL EXTEND A MINIMUM OF 6" BEYOND EQUIPMENT IN EACH DIRECTION. BOLT EQUIPMENT TO MECHANICAL SUPPORT.
- PROVIDE WITH MANUFACTURER'S CONDENSATE PUMP, OR LITTLE GIANT MINI CONDENSATE PUMP, CONCEAL PUMP BEHIND UNIT WITHIN MOUNTING BRACKET ASSEMBLY. ELECTRICAL CIRCUIT FOR PUMP SHALL BE INTEGRATED TO FAN COIL.
- ELECTRICAL TO PROVIDE DISCONNECT AND HEAT TRACE BENEATH UNIT AND TO GRAVEL BED.

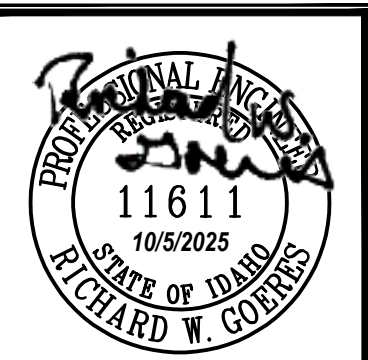
GAS-FIRED UNIT HEATER SCHEDULE

SYMBOL	AREA SERVED	UNIT TYPE	FAN			ELECTRICAL		GAS HEATING		OPERATING WEIGHT (LBS)	MANUFACTURER AND MODEL	REMARKS
			CFM	RPM	HP	V/Ø	AMPS	INPUT (MBH)	OUTPUT (MBH)			
UH-1	STORAGE / SERVICE BAY	HORIZONTAL / SEPARATED COMBUSTION	1,921	1,050	1/6	115/1	3.8	150	124.5	110	REZNOR MODEL UDXC 150	1, 2
UH-2	STORAGE / SERVICE BAY	HORIZONTAL / SEPARATED COMBUSTION	1,921	1,050	1/6	115/1	3.8	150	124.5	110	REZNOR MODEL UDXC 150	1, 2
UH-3	STORAGE / SERVICE BAY	HORIZONTAL / SEPARATED COMBUSTION	1,921	1,050	1/6	115/1	3.8	150	124.5	110	REZNOR MODEL UDXC 150	1, 2
UH-4	STORAGE / SERVICE BAY	HORIZONTAL / SEPARATED COMBUSTION	1,921	1,050	1/6	115/1	3.8	150	124.5	110	REZNOR MODEL UDXC 150	1, 2

REMARKS:

- APPROVED ALTERNATE MANUFACTURERS: HASTINGS, TRANE, MODINE, AND STERLING.
- PROVIDE UNIT WITH MANUAL SUMMER/WINTER SWITCH, THERMOSTAT AND RELAY KIT, 65° DOWNTURN NOZZLE WITH HORIZONTAL LOUVERS, AND 4-POINT SUSPENSION KIT.

NO.	REVISIONS	DATE				

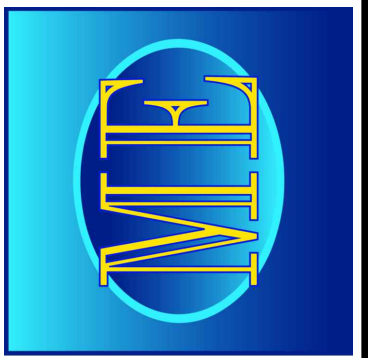


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SHED HVAC UPGRADE
BRUNEAU, ID

PROJECT	25-265
DRAWN	RG
CHECKED	CD
DATE	25-10-4
SHEET NAME	

HVAC
SCHEDULES

M3.0

These plans are approved contingent on the compliance with the mark-ups and notes applied.

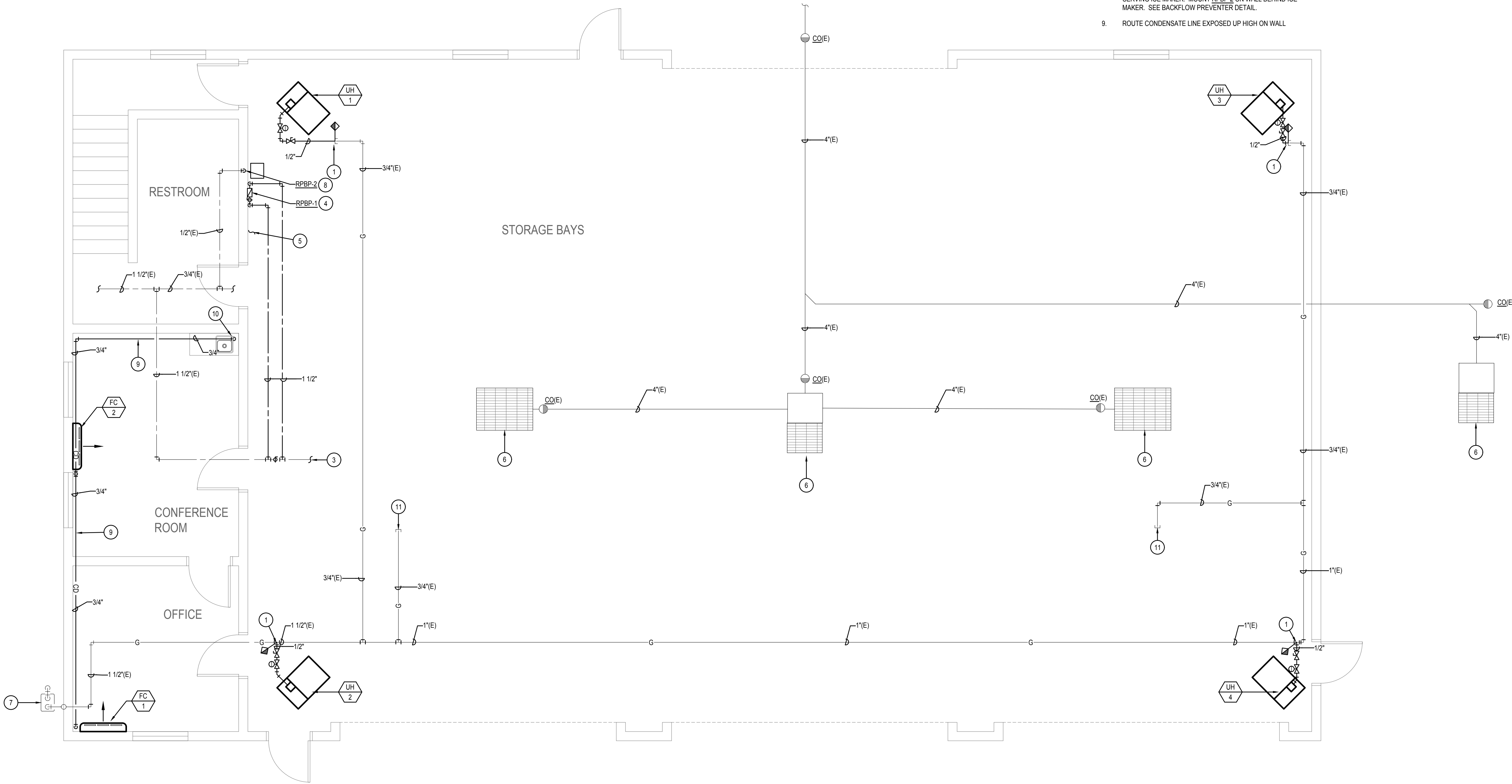
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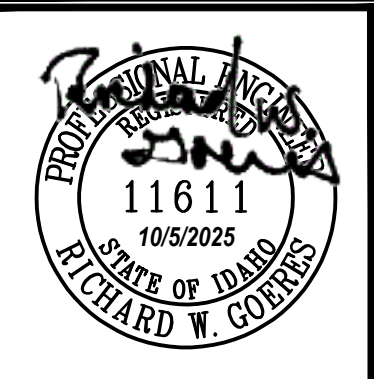
The stamped documentation has been reviewed for compliance in accordance with the 2017 ISPC as adopted by the state of Idaho by a Plumbing Plan Review. This shall not be construed as an approval of any violation of, or variance from Idaho adopted codes, laws, standards, or rules. Final approval will be based upon on-site Plumbing inspections to field verify compliance.

KEYED NOTES:

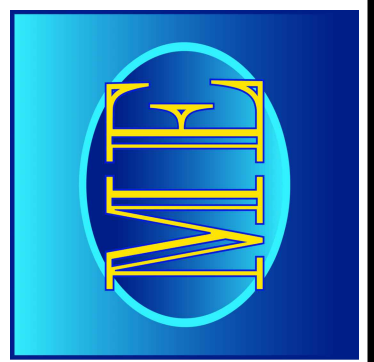
- # SYMBOL USED FOR NOTE CALLOUT.
1. EXTEND NEW GAS LINE TO UNIT HEATER FROM EXISTING LINE. PROVIDE NEW GAS COCK AND GAS REGULATOR (2 PSI - 7" W.C.) ON LINE. SEE GAS EQUIPMENT CONNECTION DETAIL.
 2. EXISTING 1/2" NON-POTABLE WATER LINE TO ICE MAKER.
 3. EXISTING 1 1/2" NON-POTABLE WATER LINE TO SHOP HOSE BIBBS.
 4. SPLICE IN REDUCED PRESSURE BACKFLOW PREVENTER INTO EXISTING WATER LINE AS SHOWN. PROVIDE BYPASS VALVE IN EXISTING LINE. SEE BACKFLOW PREVENTER DETAIL. LOCATE BACKFLOW PREVENTER ON WALL AND ROUTE BACKFLOW PREVENTER DRAIN LINE TO EXISTING DRAIN AT ICE MACHINE.
 5. ROUTE CONDENSATE LINE TO EXISTING SERVICE SINK.
 6. FLUSH AND CLEAN CATCH BASINS IN VEHICLE BAYS. USE SEWER LINE SNAKE ON ALL CATCH BASIN WASTE LINES TO CLEAR ANY BLOCKAGES.
 7. INTERMOUNTAIN GAS COMPANY TO MODIFY EXISTING GAS METER TO 2 PSI FOR NEW GAS LOAD. REFERENCE GAS SIZING CHART FOR CAPACITIES.
 8. PROVIDE NEW BACKFLOW PREVENTER ON 1/2" WATER LINE SERVING ICE MAKER. MOUNT RPBP-2 ON WALL BEHIND ICE MAKER. SEE BACKFLOW PREVENTER DETAIL.
 9. ROUTE CONDENSATE LINE EXPOSED UP HIGH ON WALL.
 10. DROP CONDENSATE LINE DOWN THROUGH COUNTERTOP AND CONNECT INTO P-TRAP AT EXISTING SINK. REPLACE EXISTING P-TRAP AT SINK WITH ONE WITH CONDENSATE DRAIN CONNECTION. SEE SINK P-TRAP DETAIL.
 11. CAPPED GAS LINE TO REMAIN.
- @ 1/8"FT. SLOPE. PAINT PIPE TO MATCH WALL.



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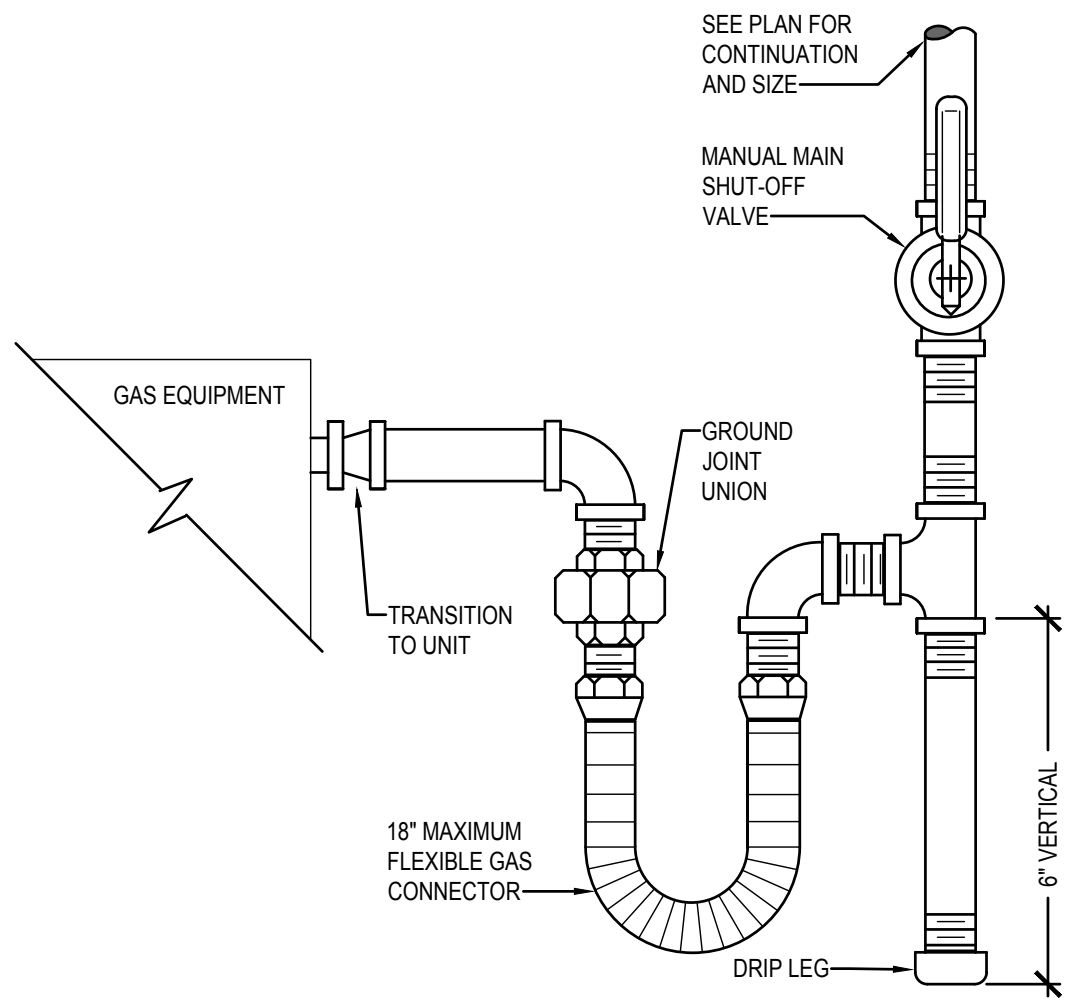
ITD D3
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BRUNEAU, ID

PROJECT	25-265
DRAWN	RG
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SHEET NAME	

PLUMBING
NEW WORK
PLAN
P1.1

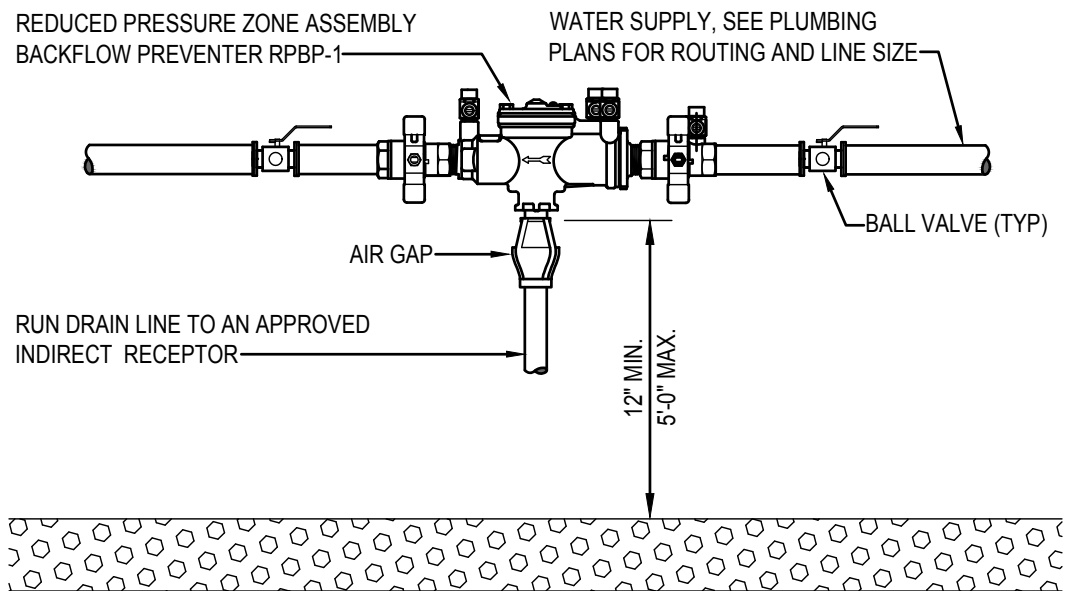
GAS SIZING CHART			
SYMBOL	INPUT (MBH)	RUNOUT SIZE (2-PSI) (INCHES)	EQUIPMENT CONNECTION SIZES (7" WC) (INCHES)
UH-1	150	1/2	1/2
UH-2	150	1/2	1/2
UH-3	150	1/2	1/2
UH-4	150	1/2	1/2
TOTAL	600	EQUIVALENT LENGTH = 200 FT PRESSURE = 2 PSI MAIN SIZE = 1 1/2"Ø	

NOTE: GAS SIZES TO EQUIPMENT ARE AS NOTED IN SCHEDULE ABOVE. ROUTE NOTED (2-PSI) GAS LINE TO GAS EQUIPMENT. PROVIDE GAS COCK AND PRESSURE REGULATOR (2-PSI TO 7" WC) SIZED FOR GAS LOAD AT EACH PIECE OF EQUIPMENT. VENT TO ATMOSPHERE PER MANUFACTURERS RECOMMENDATIONS. ROUTE NOTED (7" WC) GAS LINE TO GAS FIRED EQUIPMENT WITH GAS COCK AND FLEX CONNECTOR AT UNIT. SEE GAS CONNECTION DETAILS.



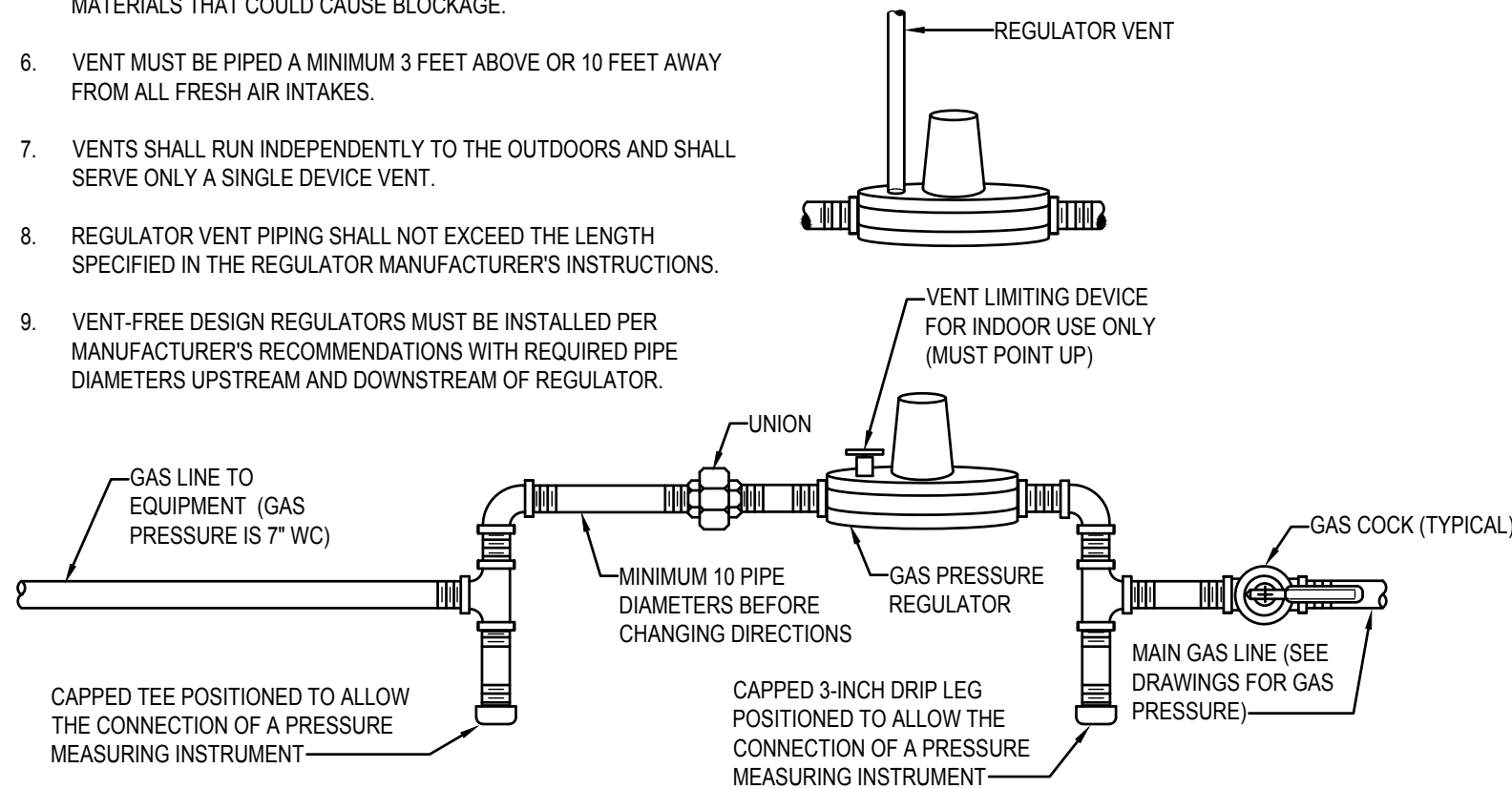
1 GAS EQUIPMENT CONNECTION DETAIL
NOT TO SCALE

PLUMBING FIXTURE SCHEDULE							
SYMBOL	FIXTURE DESCRIPTION	CONNECTION SIZE					MANUFACTURER / MODEL NUMBER / DESCRIPTION / ADDITIONAL COMMENTS
		WASTE	VENT	TRAP	CW	HW	
RPBP-1	REDUCED PRESSURE BACKFLOW PREVENTER		INDIRECT		--	--	WATTS SERIES LF009 LEAD-FREE REDUCED PRESSURE ZONE ASSEMBLY WITH QUARTER-TURN BALL VALVES, STRAINER, AND AIR GAP. CAST COPPER BODY CONSTRUCTION - 1/2" THRU 2".
RPBP-2	REDUCED PRESSURE BACKFLOW PREVENTER		INDIRECT		--	--	WATTS SERIES LF009 LEAD-FREE REDUCED PRESSURE ZONE ASSEMBLY WITH QUARTER-TURN BALL VALVES, STRAINER, AND AIR GAP. CAST COPPER BODY CONSTRUCTION - 1/2" THRU 2".

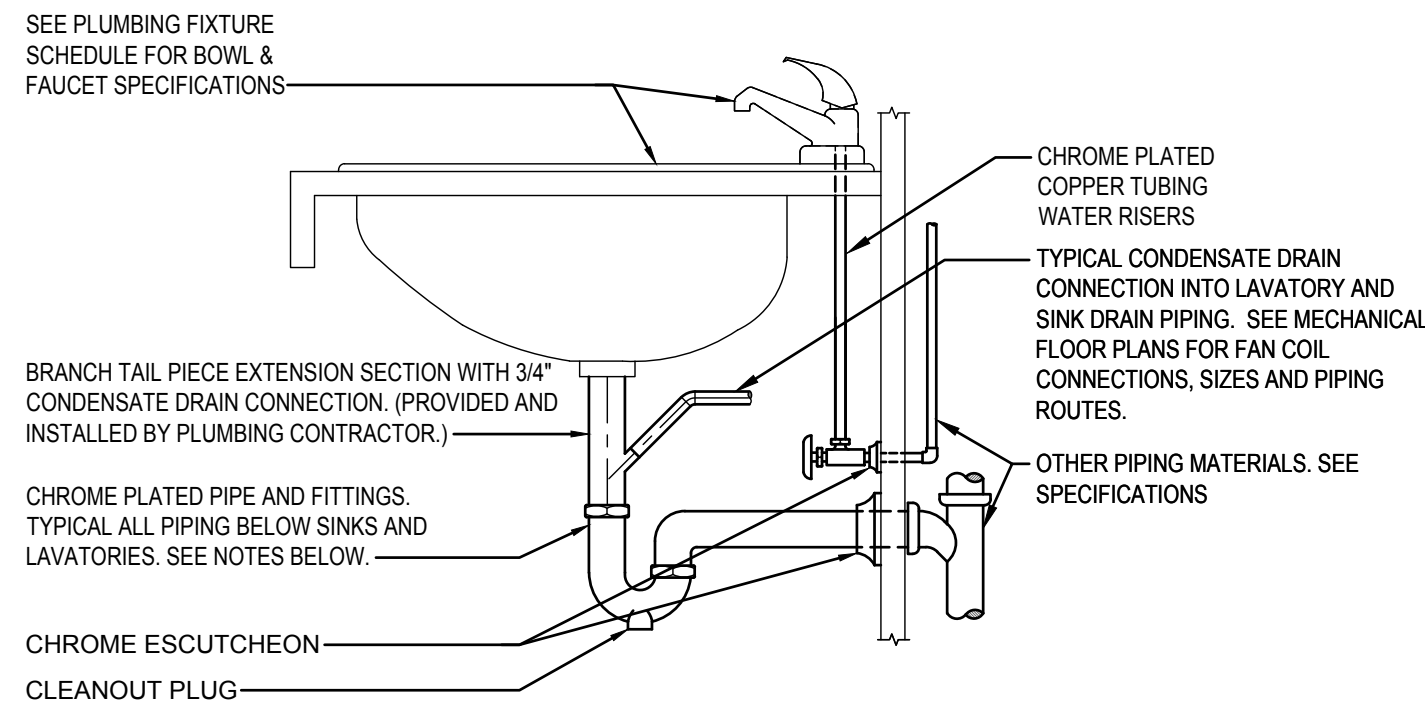


2 REDUCED PRESSURE BACKFLOW PREVENTER (RPBP)
NOT TO SCALE

- NOTES:
- VENT REGULATORS PER MANUFACTURER'S AND LOCAL GAS COMPANY'S REQUIREMENTS.
 - DO NOT REDUCE THE VENT PIPE SIZE FROM THE REGULATOR.
 - TO LIMIT THE CONSEQUENCES OF RAIN, SNOW OR DEBRIS GETTING INTO THE VENT, ALWAYS TURN THE OUTLET OF THE VENT DOWN AND ABOVE POTENTIAL WATER OR SNOW LINES.
 - PROVIDE A BUG SCREEN ON THE VENT OUTLET TO DETER INSECTS FROM NESTING IN THE LINE. NEVER PAINT OVER THE BUG SCREEN.
 - A VENT LINE PROTECTOR MAY BE USED IN OUTDOOR APPLICATIONS TO PREVENT ENTRY OF WATER, INSECTS OR OTHER FOREIGN MATERIALS THAT COULD CAUSE BLOCKAGE.
 - VENT MUST BE PIPED A MINIMUM 3 FEET ABOVE OR 10 FEET AWAY FROM ALL FRESH AIR INTAKES.
 - VENTS SHALL RUN INDEPENDENTLY TO THE OUTDOORS AND SHALL SERVE ONLY A SINGLE DEVICE VENT.
 - REGULATOR VENT PIPING SHALL NOT EXCEED THE LENGTH SPECIFIED IN THE REGULATOR MANUFACTURER'S INSTRUCTIONS.
 - VENT-FREE DESIGN REGULATORS MUST BE INSTALLED PER MANUFACTURER'S RECOMMENDATIONS WITH REQUIRED PIPE DIAMETERS UPSTREAM AND DOWNSTREAM OF REGULATOR.



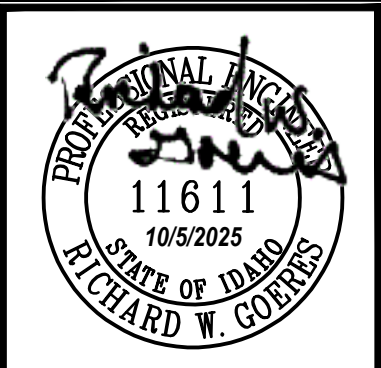
3 GAS PRESSURE REGULATOR DETAIL
NOT TO SCALE



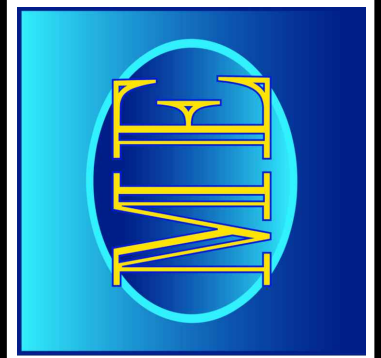
- NOTES:
- INTERIOR EXPOSED PIPE, VALVES AND FIXTURE TRIM, INCLUDING TRIM BEHIND CASEWORK DOORS, SHALL BE CHROME PLATED.
 - ALL PIPING PENETRATIONS THROUGH FINISHED WALLS SHALL BE PROVIDED WITH CHROME ESCUTCHEONS.
 - ALL SINK AND LAVATORY TRAPS SHALL BE PROVIDED WITH A CLEANOUT PLUG IN THE BOTTOM OF THE TRAP.
 - ALL PLUMBING FIXTURES SHALL BE CAULKED AND SEALED TO SURROUNDING SURFACES.
 - CONTRACTOR SHALL BE RESPONSIBLE FOR COORDINATING ALL PIPING MODIFICATIONS AND OFFSETS REQUIRED TO ACCOMMODATE SINK OR LAVATORY CASEWORK / COVERS AND ADA CLEARANCES. REFERENCE ARCHITECTURAL PLANS FOR HANDICAPPED FIXTURE DESIGNATIONS, LOCATIONS, AND MOUNTING HEIGHTS.
 - PLUMBING CONTRACTOR SHALL VERIFY THE LOCATION OF ALL LAVATORIES AND SINKS THAT NEED TO BE INSTALLED WITH THE BRANCH TAIL PIECE SECTION WITH 3/4\"/>

4 SINK/LAVATORY TAILPIECE & TRAP DETAIL
(WITH CONDENSATE DRAIN CONNECTION)
NOT TO SCALE

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**BRUNEAU MAINTENANCE
SHED HVAC UPGRADE
BRUNEAU, ID**

PROJECT	25-265
DRAWN	RG
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DATE	25-10-4
SHEET NAME	

**PLUMBING
SCHEDULES
& DETAILS**

P2.0

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ELECTRICAL LEGEND - LIGHTING

REFERENCE FIXTURE SCHEDULE FOR MOUNTING TYPE, MOUNTING HEIGHT, AND FIXTURE TYPE.

DOUBLE FACE EXIT SIGN, CEILING MOUNTED, PROVIDE UNSWITCHED CONDUCTOR.

WALL MOUNTED DOUBLE FACE EXIT SIGN PROVIDE UNSWITCHED CONDUCTOR. MOUNT AT +8'-0" UNO.

SINGLE FACE EXIT SIGN, CEILING MOUNTED PROVIDE UNSWITCHED CONDUCTOR.

WALL MOUNTED SINGLE FACE EXIT SIGN PROVIDE UNSWITCHED CONDUCTOR. MOUNT AT +8'-0" UNO.

ARROW INDICATES DIRECTION TO BE SHOWN ON SIGN.

1'X1' LIGHT FIXTURE.

1'X1' LIGHT FIXTURE, PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.

TRACK LIGHT

1'X4' LIGHT FIXTURE.

1'X4' LIGHT FIXTURE, PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.

2'X4' LIGHT FIXTURE.

2'X4' LIGHT FIXTURE, PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.

2'X2' LIGHT FIXTURE.

2'X2' LIGHT FIXTURE, PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.

DIRECT/INDIRECT LIGHT FIXTURE. SEE SCHEDULE FOR LENGTH.

DIRECT/INDIRECT LIGHT FIXTURE. SEE SCHEDULE FOR LENGTH. PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR

STRIP LIGHT FIXTURE. SEE SCHEDULE FOR LENGTH.

STRIP LIGHT FIXTURE. SEE SCHEDULE FOR LENGTH. PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR

WALL MOUNTED LIGHT FIXTURE.

WALL MOUNTED LIGHT FIXTURE, PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.

RECESSED LIGHT FIXTURE

RECESSED LIGHT FIXTURE. PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.

ROUND LIGHT FIXTURE

ROUND EMERGENCY LIGHT FIXTURE. PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.

WALL MOUNTED LIGHT FIXTURE.

WALL MOUNTED EMERGENCY LIGHT FIXTURE. PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.

POLE LIGHT 1 HEAD WITH POLE

TIME CLOCK

PHOTO CONTROL CELL LOCATED 12" ABOVE ROOF FACING NORTH.

OCCUPANCY SENSOR. PROVIDE RELAYS AND POWER PACKS AS REQUIRED.

LED DRIVER

EMERGENCY EGRESS LIGHTING WITH OUT FIXTURE HEADS. CONNECT TO AN UNSWITCHED CONDUCTOR.

EMERGENCY EGRESS LIGHTING. CONNECT TO AN UNSWITCHED CONDUCTOR.

WALL MOUNTED SINGLE FACE EXIT SIGN WITH EMERGENCY EGRESS LIGHTING. PROVIDE UNSWITCHED CONDUCTOR. MOUNT AT +8'-0" UNO.

CEILING MOUNTED. SINGLE FACE EXIT SIGN WITH EMERGENCY EGRESS LIGHTING. PROVIDE UNSWITCHED CONDUCTOR.

CEILING MOUNTED. DOUBLE FACE EXIT SIGN WITH EMERGENCY EGRESS LIGHTING. PROVIDE UNSWITCHED CONDUCTOR.

XXX INDICATES FIXTURE TYPE. REFER TO FIXTURE SCHEDULE.

EXTERIOR WALL PACK

EMERGENCY EXTERIOR WALL PACK. PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR

DEVICES

SWITCH, TYPE AS INDICATED

2 DOUBLE POLE

3 3-WAY

4 4-WAY

K KEYED

P PILOT LIGHT

D DIMMER

HP HORSEPOWER RATED

TO THERMAL OVERLOAD

LV LOW VOLTAGE

OS OCCUPANCY SENSOR

OR LOW VOLTAGE, MOMENTARY OVERRIDE

VS VACANCY SENSOR

a SUPERScript INDICATES LIGHTS TO BE SWITCHED TOGETHER

\$ DUAL LEVEL SWITCHING, INSIDE AND OUTSIDE LAMPS OF FIXTURE TO BE SWITCHED SEPARATELY.

\$ DUAL LEVEL SWITCHING WITH OCCUPANCY SENSOR, INSIDE AND OUTSIDE LAMPS OF FIXTURE TO BE SWITCHED SEPARATELY.

\$ OCCUPANCY SENSOR WITH MANUAL DIMMING, SET FOR 50% AUTOMATIC ON, AUTOMATIC OFF, WITH MANUAL DIMMING.

\$ SINGLE CONVENIENCE OUTLET

\$ FLOOR MOUNT SINGLE CONVENIENCE OUTLET

\$ DUPLEX CONVENIENCE OUTLET

\$ FLOOR MOUNT DUPLEX CONVENIENCE OUTLET

\$ EMERGENCY DUPLEX CONVENIENCE OUTLET

\$ SWITCHED DUPLEX CONVENIENCE OUTLET

\$ FLOOR MOUNTED SWITCHED DUPLEX CONVENIENCE OUTLET

\$ USB DUPLEX CONVENIENCE OUTLET

\$ USB FOURPLEX CONVENIENCE OUTLET

\$ FOURPLEX CONVENIENCE OUTLET

\$ FLOOR MOUNT FOURPLEX CONVENIENCE OUTLET

\$ CONNECTION POINT TO EQUIPMENT SPECIFIED. ELECTRICAL CONTRACTOR TO SUPPLY RACEWAY AND CONDUCTORS AND MAKE FINAL CONNECTION TO EQUIPMENT UNDER THIS SECTION. UNO

\$ FLOOR MOUNTED CONNECTION POINT. SEE NOTE ABOVE FOR REQUIREMENTS

\$ FLOOR MOUNTED JUNCTION BOX

\$ JUNCTION BOX

\$ WALL MOUNTED PUSH BUTTON, MOUNT AT SWITCH HEIGHT UNO

\$ WALL MOUNTED PUSH BUTTON, HANDICAPPED MOUNT AT SWITCH HEIGHT UNO

\$ WALL MOUNTED PUSH BUTTON, MOUNT AT SWITCH HEIGHT UNO

\$ MOTOR STARTER/CONTACTOR, SIZE/POLES NEMA 1 UNO AS INDICATED

\$ COMBINATION STARTER AND DISCONNECT, SIZE/POLES, STARTER SIZE AS INDICATED, NEMA 1 UNO

\$ FUSED DISCONNECT SWITCH, SIZE/POLES, FUSE SIZES AS INDICATED, NEMA 1 UNO

\$ NON-FUSED DISCONNECT SIZE/ POLES AS INDICATED, NEMA 1 UNO

\$ THERMOSTAT, PROVIDE CONDUIT, J-BOX, CONDUCTORS AS REQUIRED TO CONTROL ASSOCIATED UNITS. UNO COORDINATE WITH DIVISION 15.

\$ HUMIDISTAT, PROVIDE CONDUIT, J-BOX, CONDUCTORS AS REQUIRED TO CONTROL ASSOCIATED UNITS.

\$ POWER POLE - DUAL CHANNEL

\$ RECESSED ENTERTAINMENT BOX

\$ TRANSFORMER

\$ PANELBOARD. SEE SCHEDULE FOR TYPE.

\$ EQUIPMENT CABINET, SURFACE MOUNTED

\$ EQUIPMENT CABINET FLUSH MOUNTED

\$ SURFACE MULTI-OUTLET RACEWAY

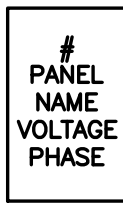
\$ MECHANICAL EQUIPMENT CALL OUT

\$ KITCHEN EQUIPMENT CALLOUT

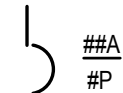
ONE LINE



DELTA WYE TRANSFORMER UNO



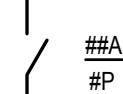
PANEL BOARD, SEE SCHEDULE FOR TYPE AND SIZE



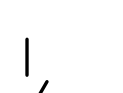
CIRCUIT BREAKER, SIZE AND POLES INDICATED



FUSE, SIZE AND TYPE INDICATED, PROVIDE FUSE FOR EACH POLE



INTERRUPTER SWITCH, SIZE AND POLES INDICATED



FUSED SWITCH, SIZE/POLES AND FUSE SIZE INDICATED



DRAW OUT CIRCUIT BREAKER, SIZE AND POLES INDICATED



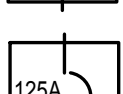
INDIVIDUAL BREAKER WITH SHUNT TRIP, SIZE AND POLES INDICATED. NEMA 1 UNO



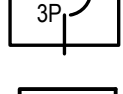
INDIVIDUAL BREAKER, SIZE AND POLES INDICATED. NEMA 1 UNO



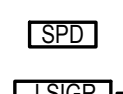
GROUND FAULT PROTECTION



SURGE SUPPRESSION DEVICE



ADJUSTABLE BREAKER SETTINGS (PER SPECIFICATIONS):
"L"-LONG TIME
"S"-SHORT TIME
"I"-INSTANTANEOUS
"G"-GROUND FAULT
"R"-ENERGY REDUCING MAINTENANCE SWITCH W/STATUS INDICATOR



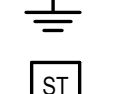
GROUND



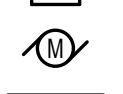
SHUNT TRIP COIL



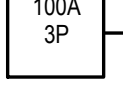
MOTOR



DISCONNECT SWITCH, SIZE AND POLES INDICATED. NEMA 1 UNO



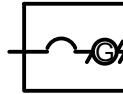
OVERHEAD SERVICE DROP



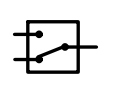
GENERATOR SET, MAIN BREAKER SIZE INDICATED



AUTOMATIC TRANSFER SWITCH (ATS)



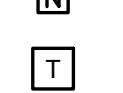
METER AND BASE



NEUTRAL



DRY TYPE TRANSFORMER



PAD MOUNT TRANSFORMER

ELECTRICAL ABBREVIATIONS

A	AMPERES
AC	6" ABOVE BACKSPLASH
AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
AF	AMP FRAME
AIC	AMPS INTERRUPTING CAPACITY
AT	AMP TRIP
ATS	AUTOMATIC TRANSFER SWITCH
AWG	AMERICAN WIRE GAUGE
BD	BOTTOM OF DECK
BS	BOTTOM OF STRUCTURE
C	CEILING MOUNTED
CB	CIRCUIT BREAKER
CF	COMPACT FLUORESCENT
CKT	CIRCUIT
CO	CONDUIT ONLY, PROVIDE PULL-LINE
CT	CURRENT TRANSFORMER
CTL	CONTROL
DC	DIRECT CURRENT
(D)	DEMOLITION
DEMO	DEMOLITION
DET	DETAIL
DTT	DOUBLE TWIN TUBE
E	EMERGENCY
(E)	EXISTING
EC	ELECTRICAL CONTRACTOR
EL	EMERGENCY LIGHT
EV	ELECTRIC VEHICLE
F	FUSE
(F)	FUTURE
FACP	FIRE ALARM CONTROL PANEL
G/GND	GROUND
GFCI	GROUND FAULT CIRCUIT INTERRUPTER
GF	GROUND FAULT INTERRUPTER
HH	HAND HOLE
HID	HIGH INTENSITY DISCHARGE
HOA	HAND-OFF-AUTO
HPS	HIGH PRESSURE SODIUM
HVAC	HEATING, VENTILATION, & AIR CONDITIONING
IG	ISOLATED GROUND
IPOCO	IDAHO POWER COMPANY
J-BOX	JUNCTION BOX
KA	KILOAMP
KVA	KILO VOLT-AMP
KW	KILOWATT
KWH	KILOWATT HOUR
LCP	LIGHTING CONTROL PANEL
MB	MAIN BREAKER
MBR	MAIN CIRCUIT BREAKER
MCC	MOTOR CONTROL CENTER
MDP	MAIN DISTRIBUTION PANEL
MLO	MAIN LUGS ONLY
MMC	MODULAR METERING CENTER
MH	METAL HALIDE
MSB	MAIN SWITCH BOARD
MTG	MOUNTING
N	NEUTRAL
(N)	NEW
NC	NORMALLY CLOSED
NEC	NATIONAL ELECTRICAL CODE
NIC	NOT IN CONTRACT
NL	NIGHT LIGHT
NO	NORMALLY OPEN
NTS	NOT TO SCALE
OH	OVERHEAD
OS	OCCUPANCY SENSOR
P	POLES
PC	PHOTO-CONTROL
PVC	POLYVINYL CHLORIDE
PWR	POWER
RE-	REFERENCE
REC	RECEPTACLE
(R)	RELOCATED
SF	SQUARE FEET
TBD	TO BE DETERMINED
TDR	TIME DELAY RELAY
TK	TOE KICK
TR	TAMPER RESISTANT
TSP	TWISTED SHIELDED PAIR
TRT	TRIPLE TUBE
TTB	TELEPHONE TERMINAL BOARD
(TYP.)	TYPICAL
UC	UNDERCABINET
UG	UNDERGROUND
U.N.O.	UNLESS NOTED OTHERWISE
V	VOLT
VA	VOLT-AMPERE
W	WATT
WG	WIRE GUARD
WP	WEATHER PROOF/NEMA 3R

PROVIDE/
PROVIDE BY
INSTALLED/
INSTALL

PROVIDE AND INSTALL / PROVIDED AND INSTALLED
BY / PROVIDE AND INSTALL

NOTE:

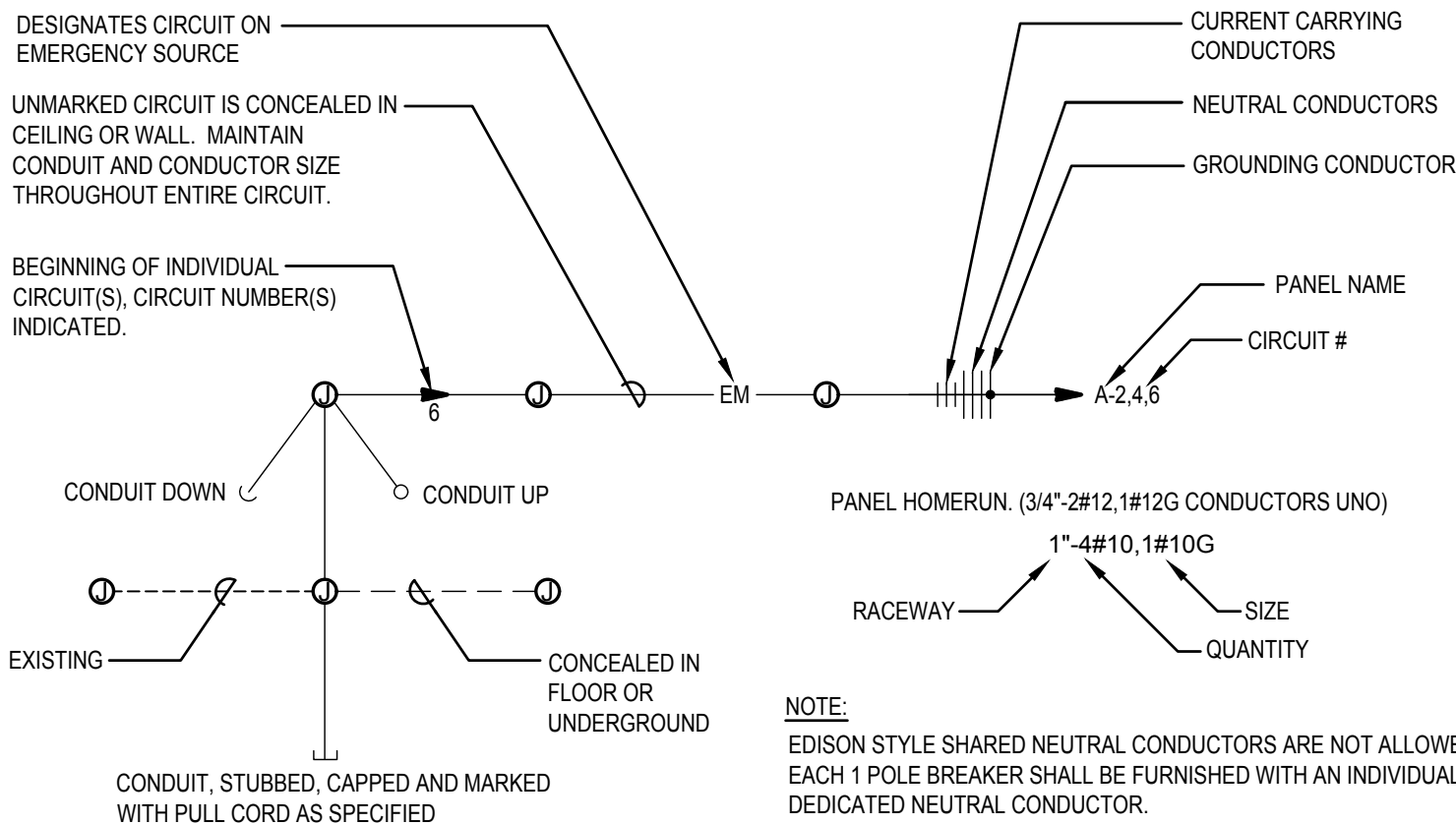
THIS IS A STANDARD LIST OF COMMONLY USED ELECTRICAL ABBREVIATIONS. SOME OF THE ABBREVIATIONS SHOWN ABOVE MAY NOT BE USED IN THIS DRAWING PACKAGE.

ELECTRICAL GENERAL NOTES

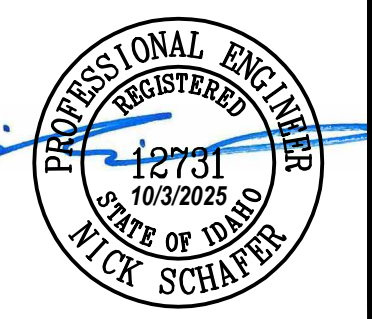
- THESE ELECTRICAL DRAWINGS ARE DIAGRAMMATIC IN NATURE; THEREFORE, THE ELECTRICAL CONTRACTOR SHALL COORDINATE ALL ELECTRICAL EQUIPMENT AND DEVICE LOCATIONS WITH ARCHITECTURAL, MECHANICAL, AND PLUMBING DIVISIONS PRIOR TO ROUGH-IN. REFER TO AND COORDINATE WITH ARCHITECT FOR EACH LOCATION WHERE SURFACE RACEWAYS ARE APPROVED, UTILIZE WIREMOLD, OR APPROVED EQUAL. SURFACE MOUNTED RACEWAYS PAINTED TO MATCH SURROUNDING WALLS.
- ALL CONDUIT AND JUNCTION BOXES ARE TO BE CONCEALED UNLESS LOCATED WITHIN DEDICATED ELECTRICAL OR MECHANICAL ROOMS. USE OF SURFACE MOUNTED RACEWAYS IN ALL OTHER SPACES MUST BE APPROVED BY THE ARCHITECT FOR EACH LOCATION WHERE SURFACE RACEWAYS ARE APPROVED, UTILIZE WIREMOLD, OR APPROVED EQUAL. SURFACE MOUNTED RACEWAYS PAINTED TO MATCH SURROUNDING WALLS.
- REFER TO THE ELECTRICAL LEGEND FOR THE DEFAULT OUTLET HEIGHT WHEN NOT INDICATED ON ELEVATIONS OR ON AT THE DEVICES.
- PROVIDE PULL-LINE IN ALL EMPTY CONDUITS.
- TERMINATE ALL LOW-VOLTAGE CONDUITS WITH INSULATED THROAT BUSHING.
- MECHANICAL EQUIPMENT INDICATED IS SHOWN IN AN APPROXIMATE LOCATION. COORDINATE EXACT LOCATION WITH MECHANICAL CONTRACTOR PRIOR TO ROUGH-IN.
- ALL NON-LOCKING, 120-V, 15 AND 20-AMP RECEPTACLES SHALL BE LISTED TAMPER-RESISTENT RECEPTACLES PER NEC 406.12
- THE ELECTRICAL DEMOLITION DRAWING(S) PROVIDED ARE INTENDED TO ASSIST THE ELECTRICAL CONTRACTOR IN ESTABLISHING AREAS REQUIRING DISCONNECTION, REMOVAL, OR RELOCATION OF ELECTRICAL EQUIPMENT, OUTLETS, WIRING, DEVICES, FIXTURES, ETC. AND MAY NOT INDICATE ALL DEVICES OR THE FULL EXTENT OF DEMOLITION AND RECONNECTION WHICH MAY BE REQUIRED. THE ELECTRICAL CONTRACTOR SHALL VISIT THE JOB SITE AND THOROUGHLY EXAMINE ALL REQUIRED DEMOLITION WORK AND INCLUDE ALL LABOR AND INCIDENTALS THAT WILL BE NECESSARY TO PERFORM DEMOLITION RECONNECTION AND TEMPORARY POWER CONNECTIONS IN THE BID.
- ALL ELECTRICAL DEVICES AND WALLS INDICATED ON THE ELECTRICAL DEMOLITION DRAWING(S) ARE TO REMAIN UNLESS OTHERWISE NOTED.

NOTE: THIS IS A STANDARD LIST OF COMMONLY USED ELECTRICAL SYMBOLS. SOME OF THE SYMBOLS SHOWN MAY NOT HAVE BEEN USED IN THIS DRAWING PACKAGE.

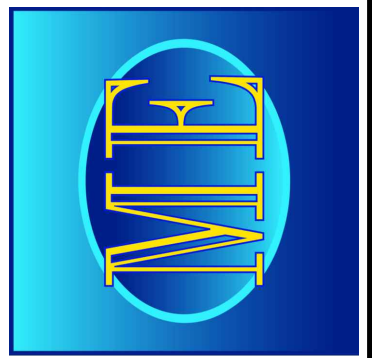
CIRCUITING SYMBOLS



DATE					
REVISIONS					
NO.					



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OVER 45 YEARS OF EXCELLENCE



ITD D3
BRUNEAU MAINTENANCE
SHED HVAC UPGRADE
BRUNEAU, ID

PROJECT	25-265
DRAWN	JRR
CHECKED	NS
DATE	25-10-4
SHEET NAME	

ELECTRICAL
COVER
SHEET

E0.0

Approved

State of Idaho
DOPL

PA#: 1906305
Date: 11/07/2025

These plans are approved contingent on the compliance with the mark-ups and notes applied.

This approval shall not be construed to be an approval of any violation of, or variance from, Idaho's adopted codes, standards, laws or rules applicable to this project.



DIVISION OF OCCUPATIONAL & PROFESSIONAL LICENSES

**COMcheck Software Version 4.1.5.5**

Interior Lighting Compliance Certificate

Project Information

Energy Code:2018 IECC

Project Title:

Project Type:New Construction

Construction Site:

Owner/Agent:

Designer/Contractor:
Musgrove Engineering
234 S. Whisperwood Way
Boise, ID 83709
208-384-0585

Additional Efficiency Package(s)

Credits: 1.0 Required 0.0 Proposed

Allowed Interior Lighting Power

A Area Category	B Floor Area (ft2)	C Allowed Watts / ft2	D Allowed Watts (B X C)
1-Workshop	1160	0.90	1044
Total Allowed Watts =			1044

Proposed Interior Lighting Power

A Fixture ID : Description / Lamp / Wattage Per Lamp / Ballast	B Lamps/ Fixture	C # of Fixtures	D Fixture Watt.	E (C X D)
1-Workshop				
LED 1: BL1: Other:	1	6	28	168
LED 2: SL1: Other:	1	4	31	124
LED 3: WB1: Other:	1	1	21	21
Total Proposed Watts =				313

Interior Lighting PASSES: Design 70% better than code

Interior Lighting Compliance Statement
Compliance Statement: The proposed interior lighting design represented in this document is consistent with the building plans, specifications, and other calculations submitted with this permit application. The proposed interior lighting systems have been designed to meet the 2018 IECC requirements in COMcheck Version 4.1.5.5 and to comply with any applicable mandatory requirements listed in the Inspection Checklist.

NICK SCHAFFER, P.E.



10/03/25

Name - TitleSignatureDate

Project Title:

Data filename: P:\Files\2025\25265\CALCS\ELEC\25265 Electrical_Compliance.cck

Report date: 10/03/25

Page 1 of 6

ENERGY CODE COMMISSIONING COMPLIANCE NOTES

SECTION 408 SYSTEM COMMISSIONING

IT SHALL BE THE ELECTRICAL CONTRACTOR'S RESPONSIBILITY TO PROVIDE ALL BELOW NOTED DOCUMENTS WITHIN 90 DAYS OF CERTIFICATE OF OCCUPANCY:

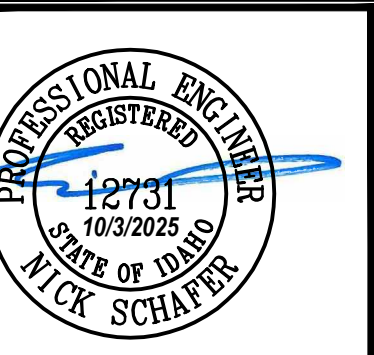
- A. AS-BUILT DRAWINGS - DRAWINGS SHALL INCLUDE THE LOCATION AND PERFORMANCE DATA OF ALL PIECES OF MECHANICAL EQUIPMENT.
- B. OPERATING AND MAINTENANCE MANUALS - MANUALS SHALL INCLUDE THE FOLLOWING:
- SUBMITTAL DATA ON ALL PIECES OF EQUIPMENT REQUIRING MAINTENANCE.
 - MANUFACTURER'S OPERATIONS AND MAINTENANCE DATA ON ALL PIECES OF EQUIPMENT. ROUTINE MAINTENANCE ACTIONS SHALL BE CLEARLY IDENTIFIED.
 - NAME AND ADDRESS AND PHONE NUMBER OF OF AT LEAST ONE (1) SERVICE PROVIDED.
 - LIGHTING CONTROL SYSTEMS MAINTENANCE AND CALIBRATION INFORMATION INCLUDING WIRING DIAGRAMS, EQUIPMENT AND SYSTEM SCHEMATICS, AND CONTROL SEQUENCES OF OPERATIONS. DESIRED OR FIELD DETERMINED SETPOINTS SHALL BE PERMANENTLY RECORDED ON CONTROL DRAWINGS AT ALL CONTROL DEVICES, OR FOR DIGITAL CONTROL SYSTEMS, IN THE SYSTEM PROGRAMMING INSTRUCTIONS.
 - A NARRATIVE ON HOW EACH LIGHTING SYSTEM IN INTENDED TO OPERATE, INCLUDING RECOMMENDED SETPOINTS.
- C. LIGHTING SYSTEM FUNCTIONAL TESTING REQUIREMENTS

FUNCTIONAL TESTING - ALL AUTOMATIC LIGHTING CONTROL SYSTEM SHALL BE FULLY TESTED TO ENSURE THE CONTROL HARDWARE AND SOFTWARE ARE CALIBRATED, ADJUSTED, PROGRAMMED, AND IN PROPER WORKING CONDITION IN ACCORDANCE WITH THE CONSTRUCTION DOCUMENTS AND MANUFACTURER'S INSTALLATION INSTRUCTIONS.

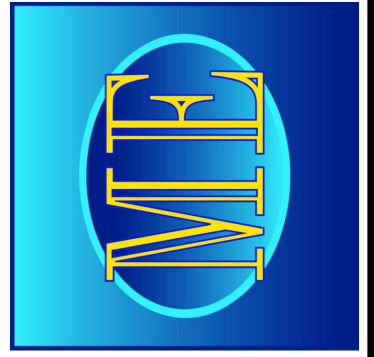
WHERE OCCUPANT SENSORS, TIME SWITCHES, PROGRAMMABLE CONTROLS, PHOTOSENSORS OR DAYLIGHTING CONTROLS ARE INSTALLED, THE FOLLOWING PROCEDURES SHALL BE PERFORMED:

- CONFIRM THAT THE PLACEMENT, SENSITIVITY AND TIME-OUT ADJUSTMENTS FOR OCCUPANT SENSORS YIELD ACCEPTABLE PERFORMANCE.
 - CONFIRM THAT THE TIME SWITCHES AND PROGRAMMABLE SCHEDULE CONTROLS ARE PROGRAMMED TO TURN THE LIGHTS OFF.
 - CONFIRM THAT THE PLACEMENT AND SENSITIVITY ADJUSTMENTS FOR PHOTOSENSOR CONTROLS REDUCE ELECTRIC LIGHT BASED ON THE AMOUNT OF USABLE DAYLIGHT IN THE SPACE AS SPECIFIED.
- D. FINAL LIGHTING SYSTEM FUNCTIONAL REPORT - A REPORT OF TEST PROCEDURES AND RESULTS IDENTIFIED AS THE "FINAL LIGHTING CONTROL REPORT" SHALL BE DELIVERED TO THE BUILDING OWNER. THE REPORT SHALL INCLUDE THE FOLLOWING:
- LIST OF FUNCTIONAL TESTS USED DURING THE COMMISSIONING PROCESS ON EACH PIECE OF EQUIPMENT.
 - RESULTS OF ALL FUNCTIONAL TESTS ON ALL PIECES OF EQUIPMENT.
 - LIST OF DEFICIENCIES FOUND AND CORRESPONDING CORRECTIVE MEASURES EITHER IMPLEMENTED OR PROPOSED ON EACH PIECE OF EQUIPMENT.
 - LIST OF EQUIPMENT NOT ABLE TO BE FUNCTIONALLY TESTED DUE TO CURRENT CLIMATE CONDITIONS. THESE PIECES OF EQUIPMENT WILL FUNCTIONALLY TESTED ONCE CLIMATE CHANGES ALLOW.

NO.	REVISIONS	DATE



MUSGROVE ENGINEERING, P.A.
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1135 Pier View Dr. #160
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ITD D3
BRUNEAU MAINTENANCE
SHED HVAC UPGRADE
BRUNEAU, ID

PROJECT	25-265
DRAWN	JRR
CHECKED	NS
DATE	25-10-4
SHEET NAME	

LIGHTING COMPLIANCE

E0.1

These plans are approved contingent on the compliance with the mark-ups and notes applied.

This approval shall not be construed to be an approval of any violation of, or variance from, Idaho's adopted codes, standards, laws, rules and regulations to this project.

DIVISION OF OCCUPATIONAL & PROFESSIONAL LICENSING



PART 1 - GENERAL
1.1 RELATED DOCUMENTS
A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Specification Sections, apply to this Section.
1.2 SUMMARY
A. Section Includes
1. Electrical, communications, controls for HVAC systems, and other related systems.
B. RDP, Registered Design Professional
C. Systems, Subsystems, and Equipment: Where these terms are used together or separately, they shall mean "as-built" systems, subsystems, and equipment.
1.4 COMMISSIONING DOCUMENTATION
A. Commissioning Plan: A commissioning plan will be developed by a registered design professional or approved agency and shall include the following items:
1. A narrative description of the activities that will be accomplished during each phase of commissioning, including the personnel intended to accomplish each of the activities.
2. A listing of the specific equipment, appliances or systems to be tested and a description of the tests to be performed.
3. Functions to be tested.
4. Conditions under which the test will be performed.
5. Measurable criteria for performance.
B. Test Checklists: RDP, with assistance of Architect/Engineer, shall develop test checklists for each system, subsystem, or equipment including interfaces and interlocks, and include a separate entry, with space for comments, for each item to be tested. Prepare separate checklists for each mode of operation and provide space to indicate whether the mode under test responded as required. Provide space for testing personnel to sign off on each checklist.
1. Name and identification of tested item.
2. Test number.
3. Time and date of test.
4. Indication of whether the record is for a first test or retest following correction of a problem or issue.
5. Date of the test and name of parties involved as applicable.
6. Individuals present for test.
7. Deficiencies/issues/Results of test.
8. Note if re-test is necessary.
C. Test and Inspection Reports: RDP shall record test data, observations, and measurements on test checklists. Photographs, forms, and other means appropriate for the application shall be included with data. RDP shall compile test and inspection reports and tests and inspection certificates and include them in systems manual and commissioning report.
D. Corrective Action Documents: RDP shall document corrective action taken for systems and equipment that fail tests. Include required modifications to systems and equipment and revisions to test procedures, if any. Retest systems and equipment requiring corrective action and document retest results.
E. Issues Log: RDP shall prepare and maintain an issues log that describes problems, installation, and performance issues that are at variance with the Contract Documents. Identify and track issues as they are encountered, documenting the status of unresolved and resolved issues.
1. Creating an Issues Log Entry:
a. Identify the issue with unique numeric or alphanumeric identifier by which the issue may be tracked.
b. Assign a descriptive title of the issue.
c. Identify date and time of the issue.
d. Identify last number of being performed at the time of the observation, if applicable, for cross-reference.
e. Identify system, subsystem, and equipment to which the issue applies.
f. Identify location of system, subsystem, and equipment.
g. Include information that may be helpful in diagnosing or evaluating the issue.
h. Note recommended corrective action.
i. Identify commissioning team member responsible for corrective action.
j. Identify expected date of correction.
k. Identify person documenting the issue.
2. Documenting Issue Resolution:
a. Log date correction is completed or the issue is resolved.
b. Describe corrective action or resolution taken. Include description of diagnostic steps taken to determine root cause of the issue, if any.
c. Identify changes to the Contract Documents that may require action.
d. State that correction was completed and system, subsystem, and equipment is ready for retest, if applicable.
e. Identify person(s) who corrected or resolved the issue.
3. Issues Log Report: On a periodic basis, but not less than for each commissioning team meeting, RDP shall prepare a written narrative for review of outstanding issues and a status update of the issues log. As a minimum, RDP shall include the following information in the issues log and expand it in the narrative:
a. Issue number and title.
b. Date of the identification of the issue.
c. Name of the commissioning team member assigned responsibility for resolution.
d. Expected date of correction.
F. Commissioning Report: RDP shall document results of the commissioning process including unresolved issues and performance of systems, subsystems, and equipment. The commissioning report shall indicate whether systems, subsystems, and equipment have been completed and are performing according to the Contract Documents. The commissioning report shall include, but is not limited to, the following:
1. Lists and explanations of substitutions, compromises, variances in the Contract Documents, record of conditions, and, if appropriate, recommendations for resolution. This report shall be used to evaluate systems, subsystems, and equipment and shall serve as a future reference document during Owner occupancy and operation. It shall describe components and performance that exceed requirements of the Contract Documents. It may also include a recommendation for accepting or rejecting systems, subsystems, and equipment.
2. Commissioning plan.
3. Testing plans and reports.
4. Corrective modification documentation.
5. Issues log.
6. Completed test checklists.
G. Systems Manual: RDP shall gather required information and compile systems manual. Systems manual shall include, but is not limited to, the following:
1. Submittal Data stating equipment installed and selected options for each piece of equipment requiring maintenance.
2. Operation and maintenance data on each piece of equipment requiring maintenance. Required routine maintenance actions shall be clearly identified.
3. Name and address of at least one service agency.
4. Lighting controls system maintenance and calibration information.
5. A narrative of how each system is intended to operate, including recommended setpoints.

PART 2 - PRODUCTS
PART 3 - EXECUTION
3.1 TESTING
A. Testing shall ensure that the control hardware and software are calibrated, adjusted, programmed and in proper working condition in accordance with the construction documents and manufacturers installation instructions.
B. Testing shall ensure that the lighting controls meet all provisions of the applicable energy code.
C. Perform tests using design conditions whenever possible. Where occupant sensors, time switches, programmable schedule control, photosensors or daylighting controls are installed, the following procedures shall be performed:
1. Confirm that the placement, sensitivity and time-out adjustments for occupant sensors yield acceptable performance.
2. Confirm that the time switches and programmable schedule controls are programmed to turn the lights off.
3. Confirm that the placement and sensitivity adjustments for photosensor controls reduce electric light based on the amount of usable daylight in the space as specified.
END OF SECTION 260800
SECTION 260923 - LIGHTING CONTROL DEVICES
PART 1 - GENERAL
1.1 RELATED DOCUMENTS
A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.
1.2 SUMMARY
A. This Section includes time switches, photoelectric relays, occupancy sensors, and multi-pole lighting relays and controllers.
B. Related Sections include the following:
1. Section 262726 - Wiring Devices for wall-box dimmers and manual light switches.
1.3 SUBMITTALS
A. Submit shop drawings and product data, including all wiring diagrams.
PART 2 - PRODUCTS
2.1 GENERAL LIGHTING CONTROL DEVICE REQUIREMENTS
A. Line-Voltage Surge Protection: Include in all 120- and 277-V solid-state equipment. Comply with UL 1449.
2.2 LIGHTING CONTROL SYSTEM
A. Description of Work: Extent of lighting control system work is indicated by drawings, and by the requirements of this section. It is defined to include low voltage lighting control panels, switch inputs, and wiring.
B. SYSTEM DESCRIPTION
1. The lighting controls are based on the Lutron Vibe system. No substitutions.
2. Refer to the plans for additional information.
3. Provide Vibe Hubs as required for a complete system.
C. MANUFACTURERS
A. Lutron
D. PRODUCT SUPPORT AND SERVICE
1. Factory Support: Factory telephone support shall be available at no cost to the owner. Factory assistance shall consist of solving programming or application questions concerning the control equipment.
E. WARRANTY
A. Manufacturer shall supply a 2 year warranty on all hardware and software.
PART 3 - EXECUTION
3.1 INSTALLATION
A. Install equipment level and plumb and according to manufacturer's written instructions.
3.2 CONTROL WIRING INSTALLATION
A. Install wiring between sensing and control devices according to manufacturer's written instructions.
B. Wiring Method: Install all wiring in raceways.
C. Bundle, train, and support wiring in enclosures.
D. Ground equipment.
E. Connections: Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values.
3.3 IDENTIFICATION
A. Provide Brady wire markers or equivalents on all conductors.
3.4 FIELD QUALITY CONTROL
A. Inspect control components for defects and physical damage.
B. Verify settings of photoelectric devices with photosensor.
C. Electrical Tests: Use particular caution when testing devices containing solid-state components. Perform the following according to manufacturer's written instructions:
1. Continuity tests of circuits.
2. Operational Tests: Set and operate devices to demonstrate their functions and capabilities in a methodical sequence that cues and reproduces actual operating functions.
D. Correct deficiencies, make necessary adjustments, and retest. Verify that specified requirements are met.
CLEANING
A. Cleaning: Clean equipment and devices internally and externally using methods and materials recommended by manufacturers, and repair damaged finishes.
END OF SECTION 260923
SECTION 262416 - PANELBOARDS
PART 1 - GENERAL
1.1 RELATED DOCUMENTS
A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.
1.2 SUMMARY
A. This Section includes load centers and panelboards, overcurrent protective devices, and associated auxiliary equipment rated 600 V and less for the following types:
1. Lighting and appliance branch-circuit panelboards.
2. Distribution panelboards.
B. Related sections:
1. Section 260501 - Field Test and Operational Check.
2. Section 260526 - Grounding.
1.3 SUBMITTALS
A. Product Data: For each type of panelboard, overcurrent protective device, accessory, and component indicated. Include dimensions and manufacturers' technical data on features, performance, electrical characteristics, ratings, and finishes.

B. Shop Drawings: For each panelboard and related equipment.
1. Dimensioned plans, elevations, sections, and details. Show tabulations of installed devices, equipment features, and ratings. Include the following:
a. Enclosure types and details for types other than NEMA 250, Type 1.
b. Bus configuration, current, and voltage ratings.
c. Short-circuit current rating of panelboards and overcurrent protective devices.
d. UL listing for series rating of installed devices.
e. Features, change in design, ratings, and factory settings of individual overcurrent protective devices and auxiliary components.
C. Panelboard Schedules: For installation in panelboards. Submit final versions after load balancing.
D. Maintenance Data: For panelboards and components, include the following:
1. Manufacturer's written instructions for testing and adjusting overcurrent protective devices.
2. Time-current curves, including selective ranges for each type of overcurrent protective device.
1.4 QUALITY ASSURANCE
A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in the NEC, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
B. Comply with NEMA PB 1.
C. Comply with the NEC.
1.5 COORDINATION
A. Coordinate layout and installation of panelboards and components with other construction that penetrates walls or is supported by them, including electrical and other types of equipment, raceways, piping, and encumbrances to workspace clearance requirements.
PART 2 - PRODUCTS
2.1 MANUFACTURERS
A. Manufacturers:
1. Panelboards, Overcurrent Protective Devices and Accessories:
a. Eaton Corp., Cutler-Hammer Products.
b. General Electric Co.: Electrical Distribution & Control Div.
c. Siemens.
d. Square D Co.; Schneider Electric Brands.
e. Or approved equal.
2.2 FABRICATION AND FEATURES
A. Enclosures: Flush or surface mounted cabinets (as indicated on drawings). Construct cabinets with code gauge galvanized steel. Provide minimum 20" wide cabinets and extra wiring space where incoming feed-through or parallel lines are shown. NEMA PB 1, Type 1, to meet environmental conditions at installed location.
1. Outdoor Locations: NEMA 250, Type 3R.
2. Kitchen Areas: NEMA 250, Type 4X, stainless steel.
3. Other Wet or Damp Indoor Locations: NEMA 250, Type 4.
4. Hazardous Areas Indicated on Drawings: NEMA 250, Type 7C.
B. Front: Secured to box with concealed trim clamps. For surface-mounted fronts, match box dimensions; for flush-mounted fronts, overlap box.
C. Doors: Provide door-in door construction, mild-rolled steel. Inner door shall provide access to breaker handles and outer door shall provide access to wiring space as well. Inner door shall be completely flush with no visible bolts, screw-heads or hinges and with flush catch and lock. Outer door shall have concealed hinges, flush catch and lock to match inner door, located in line with inner door catch. (Tee bar handles are not acceptable).
D. Finish: Manufacturer's standard enamel finish over corrosion-resistant treatment or primer coat.
E. Directory Card: With transparent protective cover, mounted inside metal frame, inside panelboard door.
F. Bus: Hard-drawn copper, 98 percent conductivity. Attach circuit breakers to bus so that circuits 1, 3, and 5; 2, 4, and 6, or any three similarly numbered circuits form one three-phase, four-wire circuit.
G. Main and Neutral Lugs: Compression or mechanical type suitable for use with conductor material.
H. Equipment Ground Bus: Adequate for feeder and branch-circuit equipment ground conductors, bonded to box.
I. Future Devices: Mounting brackets, bus connections, and necessary appliances required for future installation of devices.
J. Isolated Equipment Ground Bus: Where indicated on drawings - Adequate for branch-circuit equipment ground conductors; insulated from box.
K. Extra-Capacity Neutral Bus: Where indicated on drawings, neutral bus rated 200 percent phase bus and UL listed as suitable for nonlinear loads. Where indicated on plans, On 120/208V Panels fed by 4 factor Types Transformer.
L. Skirt for Surface-Mounted Panelboards: Same gauge and finish as panelboard front with flanges for attachment to panelboard, wall, and ceiling or floor. Where indicated on plans.
M. Gutter Barrier: Arrange to isolate individual panel sections.
N. Feed-through Lugs: Compression or mechanical type suitable for use with conductor material. Locate at opposite end of bus from incoming lugs or main device. For two-section panels.
O. Panels located adjacent to each other shall have identically sized enclosures and trims.
2.3 PANELBOARD SHORT-CIRCUIT RATING
A. UL listed indicating series-connected rating with integral or remote upstream devices. Include size and type of upstream device allowable, branch devices allowable, and UL series-connected short-circuit rating. If not series rated: Fully rated to interrupt symmetrical short-circuit current available at terminals or the rating indicated on the plans, whichever is higher.
2.4 LIGHTING AND APPLIANCE BRANCH-CIRCUIT PANELBOARDS
A. Branch Overcurrent Protective Devices:
1. 120/208V volt branch circuit panelboards: Quick-make, quick-break, molded case plug-in type designed for 120/208V volt, three-phase, four-wire service with minimum 14,000 amperes rms short circuit rating.
2. 277/480V volt branch circuit panelboards: Molded case bolt-on type designed for 277/480V volt, three-phase, four-wire service with minimum 14,000 amperes rms short circuit rating.
3. Provide multi-pole units with common trip elements.
4. Breaker shall have center-tripped position in addition to the ON and OFF positions.
5. Provide lockouts for all circuits that should not be inadvertently tripped (as indicated on the drawings).
2.5 OVERCURRENT PROTECTIVE DEVICES
A. Feeder Circuit Breaker Assemblies 400 Amps or larger:
1. Feeder Circuit breakers 400 Amps or larger shall be digital solid state true RMS sensing Molded Case Circuit Breakers with temperature insensitive, solid state trips, current sensors and solid state logic circuit integral with the frame. All circuit breakers shall be of same design for over-current and ground fault trip coordination. The Circuit Breakers shall have the following minimum features:
a. UL listed for 80 percent load application unless otherwise indicated on plans.
b. Long time pickup (ampere setting) determined by interchangeable rating plug.
c. Adjustable instantaneous with short time tripping function.
d. Circuit Breaker shall allow the UL listed field installation internal accessories (Auxiliary Switch, Shunt Trip, Undervoltage release, Bell Alarm Switch) without removal of cover to install. Circuit Breaker shall include Accessories as indicated on plans.
e. Circuit breaker handle accessories shall provide provisions for locking handle in the 'ON' or 'OFF' position.
2. Where specifically indicated or required by NEC:
a. Adjustable [L] Long time time-delay and ampere setting.
b. Adjustable [S] Short time-delay and pick-up.
c. Adjustable [I] Instantaneous trip.
d. Adjustable [G] Ground fault pickup and delay where indicated or required by NEC.
e. Where indicated special zone control interlocking for main breaker and future main and tie breaker of double-ended substation switchboard.
f. Short circuit, overload and ground fault trip indicators.
g. Trip device of circuit breakers shall be of same type for tripping coordination.
B. Feeder Circuit Breaker Assemblies 150 Amp and below:
1. Feeder Circuit breakers 150 Amp and below shall be magnetic Magnetic Circuit Breaker. Inverse time current element for low level overloads, and instantaneous magnetic trip element for short circuit, unless otherwise indicated or required to meet Section 2.4 C above. Minimum features below:
a. UL listed for 80 percent load application unless otherwise indicated on plans.
b. Circuit Breaker shall allow the UL listed field installation internal accessories (Auxiliary Switch, Shunt Trip, Undervoltage release, Bell Alarm Switch) without removal of cover to install. Circuit Breaker shall include Accessories as indicated on plans.
c. Circuit breaker handle accessories shall provide provisions for locking handle in the 'ON' or 'OFF' position.
2. Where specifically indicated or required by NEC:
a. Adjustable [L] Long time time-delay and ampere setting with Long time pickup (ampere setting) determined by interchangeable rating plug.
b. Adjustable [S] Short time-delay and pick-up.
c. Adjustable [I] Instantaneous trip.
d. Adjustable [G] Ground fault pickup and delay where indicated or required by NEC.
e. Where indicated special zone control interlocking for main breaker and future main and tie breaker of double-ended substation switchboard.
f. Short circuit, overload and ground fault trip indicators.
g. Trip device of circuit breakers shall be of same type for tripping coordination.
C. General Breaker Requirements:
1. Minimum interrupting capacity shall match the minimum required interrupt rating of the panel.
2. Standard frame sizes, trip ratings, and number of poles.
3. Lugs: Mechanical or compression style, suitable for number, size, trip ratings, and material of conductors.
4. Application Listing: Appropriate for application; Type SVD for switching Fluorescent lighting loads, Type HACR for heating, air-conditioning, and refrigerating equipment.
5. GFCI Circuit Breakers: Single- and two-pole configurations with 5-mA trip sensitivity.
6. Shunt Trip: 120-V trip coil energized from separate circuit.
PART 3 - EXECUTION
3.1 INSTALLATION
A. Install panelboards and accessories according to NEMA PB 1.1.
B. Clearances: Minimum code required clearances around panelboards must be maintained.
C. Mounting Heights: Top of trim 78 inches above finished floor, unless otherwise indicated.
D. Mounting: Plumb and rigid without distortion of box. Mount recessed panelboards with fronts uniformly flush with wall finish.
E. Mounting Hardware: Provide all necessary blocking, channels and other hardware for securing panelboards to wall, column or other parts of building structure.
F. Circuit Directory: Create a directory to indicate installed circuit loads after balancing panelboard loads. Obtain approval before installing. Use a computer or typewriter to create directory; handwritten directories are not acceptable.
G. Install filler plates in unused spaces.
H. Wiring in Panelboard Cutters: Arrange conductors into groups and bundle and wrap with wire ties after completing load balancing.
3.2 IDENTIFICATION
A. Identify field-installed conductors, interconnecting wiring, and components.
B. Panelboard Nameplates: Label each panelboard with engraved metal or laminated-plastic nameplate mounted with corrosion-resistant screws. Label shall include panel designation, voltage and phase in minimum 1/2" high letters.
3.3 CONNECTIONS
A. Install equipment grounding connections for panelboards with ground continuity to main electrical ground bus.
B. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values.
3.4 FIELD QUALITY CONTROL
A. After installing panelboards and after electrical circuitry has been energized, demonstrate product capability and compliance with requirements.
1. Procedures: Perform each visual and mechanical inspection and electrical test indicated in NETA ATS, Section 7.6 for molded-case circuit breakers. Certify compliance with test parameters.
2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
B. Balancing Loads: After Substant Completion, measure load balancing and make circuit changes as follows:
1. Measure as directed during period of normal system loading.
2. Perform load-balancing circuit changes outside normal occupancy/working schedule of the facility and at time directed.
3. After circuit changes, recheck electric during normal load period. Record all load readings before and after changes and submit test records.
4. Tolerance: Difference exceeding 20 percent between phase loads, within a panelboard, is not acceptable. Rebalance and recheck as necessary to meet this minimum requirement.
3.5 ADJUSTING
A. Set field-adjustable switches and circuit breaker trip ranges.
B. Adjust all operating mechanisms for free mechanical movement.
CLEANING
A. On completion of installation, inspect interior and exterior of panelboards. Remove paint splatters and other spots. Vacuum dirt and debris; do not use compressed air to assist in cleaning. Repair exposed surfaces to match original finish.
END OF SECTION 262416
SECTION 262726 - WIRING DEVICES
PART 1 - GENERAL
1.1 RELATED DOCUMENTS
A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.
1.2 SUMMARY
A. This Section includes receptacles, switches, and finish plates.
1.3 DEFINITIONS
A. GFCI: Ground-fault circuit interrupter.
B. TVSS: Transient voltage surge suppressor.
1.4 SUBMITTALS
A. Submit shop drawings and product data.
1.5 COORDINATION
A. Receptacles for Owner-Furnished Equipment. Match plug configurations.
PART 2 - PRODUCTS
2.1 MANUFACTURERS
A. Manufacturers:
1. Bryant; Hubbell, Inc.
2. GE Company; GE Wiring Devices.
3. Hubbell Wiring Device - Kelco.
4. Leviton Manufacturing Co., Inc.
5. Pass & Seymour/Legrand; Wiring Devices Div.
6. Cooper Wiring Devices.
7. Or approved equal.
C. Wiring Devices for Hazardous (Classified) Locations:
1. Crouse-Hinds Electrical Co.; Distribution Equipment Div. or approved equal.
D. Multi-outlet Assemblies:
1. Wornold.
2. Hubbell, Inc.; Wiring Devices Div.
3. Or approved equal.
2.2 RECEPTACLES
A. General Requirements for All Devices
1. Each device shall have an ampere rating not less than that of the branch circuit(s) overcurrent protection device. **White color**, unless noted otherwise.
a. Emergency devices backed up by an emergency generator and the associated cover plates shall be Red color.
2. Wiring Devices, Components, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
3. All devices shall be Commercial Specification Grade (Construction specification grade is prohibited), unless noted otherwise.
a. All convenience receptacles shall be Heavy-Duty 125 V, 20 A; Comply with NEMA WD 1, NEMA WD 6 Configuration 5-20R, UL 498, and FS W-C-596. Subject to compliance with requirements, provide Hubbell Wiring Device-Kelco; 5362 Series or similar.
b. Subject to compliance with requirements, provide Hubbell Wiring Device-Kelco; 5362 Series or similar.
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gw. Subject to compliance with requirements, provide Hubbell